Network Science

Lecture 06: Structural Balance

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Speaker notes

This lecture moves from unsigned to **signed** graphs — edges carry a positive or negative label representing alliance/rivalry, trust/distrust, like/dislike. The central insight is that a simple local constraint on triangles forces a surprisingly rigid global structure. That interplay between local psychology and global organization is what makes structural balance one of the most elegant results in network science. The lecture continues the course arc by introducing a fourth kind of gap: the Balance Theorem applies to complete graphs, but real networks are sparse and noisy — so we need approximate balance measures, the same modeling/measurement pattern from L03–L05.

From Unsigned to Signed Graphs

So far, an edge meant "connected." Starting now, every edge also carries a **sign**: positive (+) for alliance, friendship, or trust; negative (−) for rivalry, hostility, or distrust.

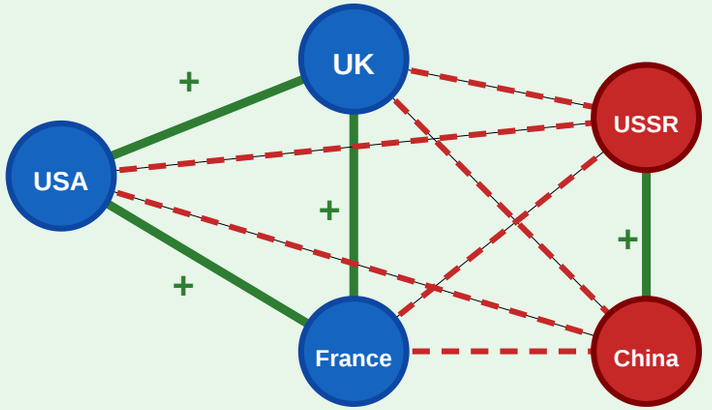
This one addition — a single bit per edge — turns out to constrain global structure powerfully.

A Motivating Scenario

Cold War alliances as a signed graph

green solid = alliance (+) · red dashed = rivalry (−)

Pre-1960 — two balanced blocs



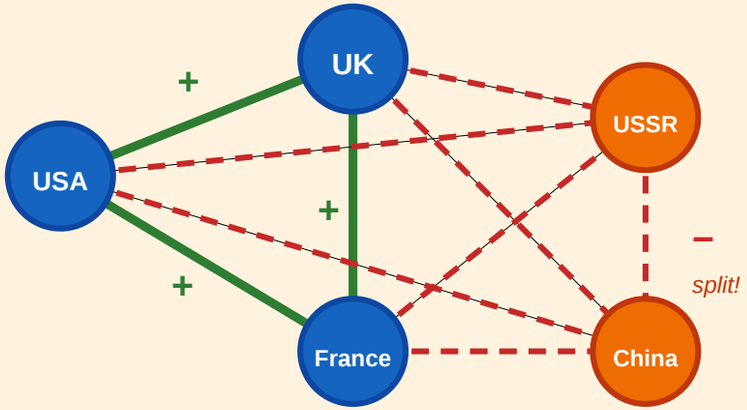
Western Bloc | **Eastern Bloc**

All 10 triangles are balanced

→ Balance Theorem: exactly 2 camps

✓ *strong structural balance*

Post-1969 — Sino-Soviet split



△ USA–USSR–China: all negative (−−−)

→ violates strong balance

→ but satisfies weak balance (≥3 camps)

camps: {USA,UK,FR} · {USSR} · {China}

Five nations during the Cold War. **Left:** pre-1960, all alliances and rivalries form two clean blocs — every triangle is balanced. **Right:** after the 1969 Sino-Soviet split, USSR–China flips to −, creating all-negative triangles. The two-bloc structure breaks.

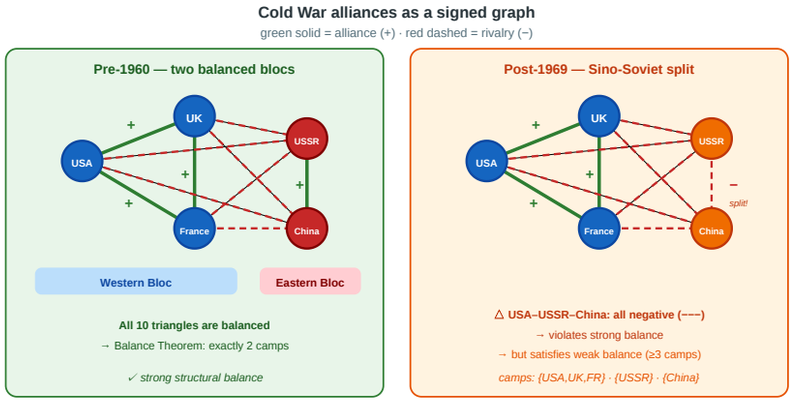
Speaker notes

This figure is the anchor for the lecture. Use it to seed the four questions on the next slide. By the end of the session students should be able to (a) classify every triangle in both panels, (b) explain why the left panel is balanced and the right is not under strong balance, (c) show that the right panel satisfies weak balance with three coalitions, and (d) connect the triangle counts to Leskovec et al.'s empirical findings.

Four Questions from One Picture

1. **Local stability.** Which signed triangles feel psychologically "stable" — and which feel like they should change?
2. **Global structure.** If every triangle is stable, what does that force the *whole* graph to look like?
3. **Relaxation.** When a three-way rivalry (all −) appears, does the two-bloc theorem still hold — or do we need more camps?
4. **Measurement.** In a real signed network with millions of edges, how close to balanced is it — and how would we know?

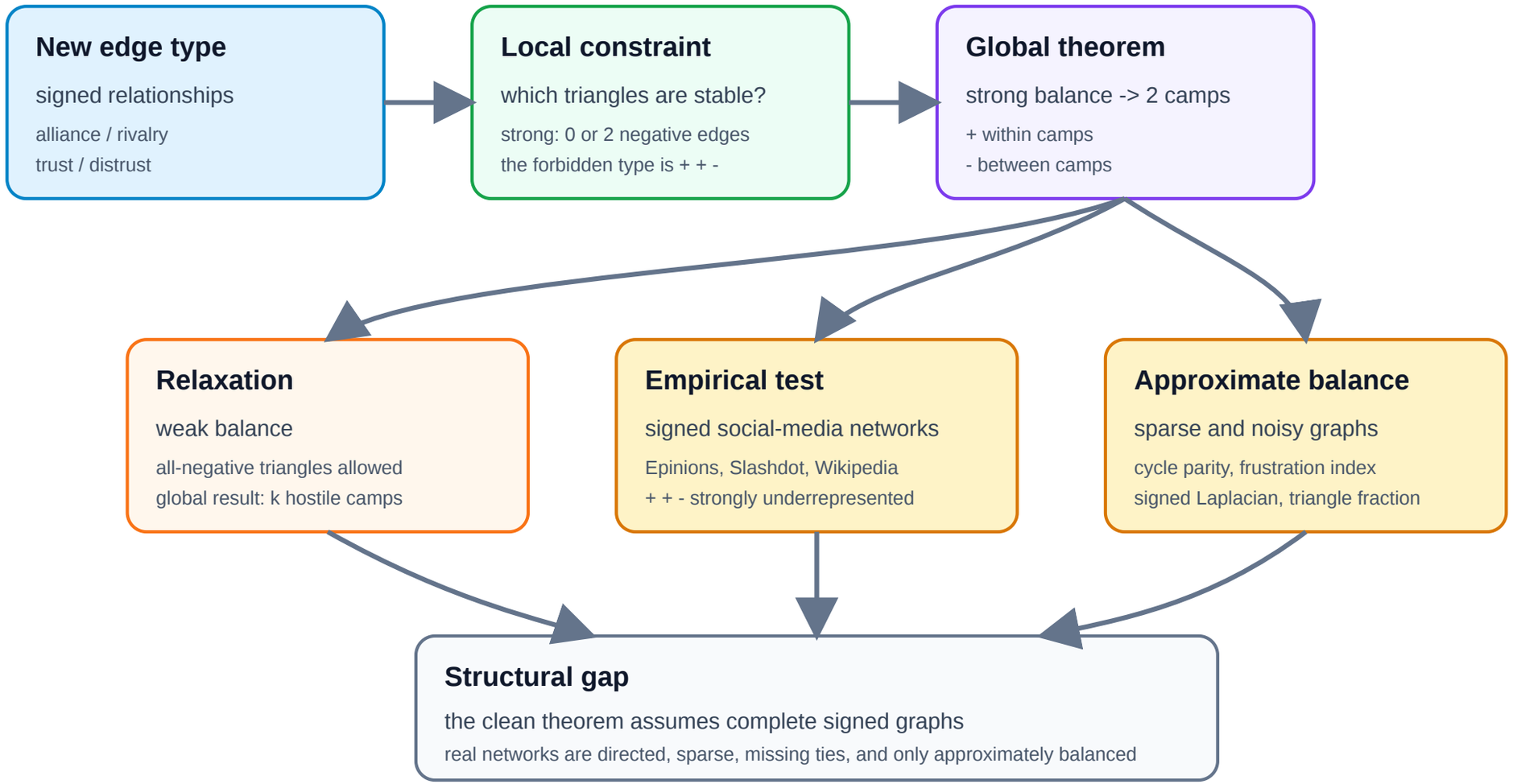
Every concept in today's lecture answers one of these.



Speaker notes

Questions 1–2 are Module 1 (Balance Theorem). Question 3 is Module 2 (Weak Balance). Question 4 spans Modules 3–4 (empirical test and approximate balance). Keep referring back to the Cold War figure throughout.

Argument Map



Speaker notes

This map gives the lecture's logical route. Adding signs to edges creates a local triangle classification problem; imposing the local rule on a complete graph yields the strong balance theorem and the two-camp prediction. Weak balance relaxes the all-negative triangle and gives k camps. The empirical part then asks whether signed online networks show the forbidden triangle patterns, and the final module asks how balance can be measured when the theorem's complete-graph assumptions fail.

A New Kind of Gap

L03–L04 had a **computational** gap: MaxSTC and modularity maximization are NP-hard. L05 had a **causal** gap: cross-sectional data cannot identify selection vs. socialization.

L06 has a **structural** gap: the Balance Theorem requires a *complete* signed graph, but real networks are *sparse*. Most pairs have no observed edge, so the theorem's premise is never met exactly.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance measures

Speaker notes

Naming the gap type up front gives students a framework for understanding why the clean theorems of Module 1 do not directly apply to real data, and why Modules 3–4 are necessary.

Takeaway

> Adding a sign to each edge creates a surprisingly powerful local constraint. The structural balance theorem turns triangle-level psychology into a global two-bloc prediction — but applying it to real data requires dealing with incompleteness and approximation.

Signed Graphs and Structural Balance

Guiding Question: If every triangle in a signed graph is "psychologically stable," what does that force the global structure to be?

Speaker notes

This module introduces signed graphs formally, presents Heider's psychological motivation, classifies the four triangle types, states the Balance Theorem, and sketches its proof. It answers Questions 1 and 2 from the opening.

Heider's Insight (1946)

Fritz Heider (1946), *Attitudes and Cognitive Organization*, **Journal of Psychology 21, 107–112**.

Heider proposed that people experience psychological tension ("cognitive dissonance") when their relationships form certain patterns — for example, two friends who both dislike the same person feel stable, but two friends who disagree about a third person feel pressure to change.

Speaker notes

Heider's paper is the origin of the entire field. His psychological intuition — that certain configurations of friend/enemy relationships are stable and others are unstable — was formalized into a graph-theoretic result by Cartwright & Harary (1956). The result is one of the earliest examples of a local psychological rule producing a provable global structural constraint.

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The network-science formalization: encode the tension as a constraint on **signed triangles**.

Signed Graphs — Formal Definition

A **signed graph** is a pair (G, σ) where $G = (V, E)$ is a graph and $\sigma : E \to \{+, -\}$ assigns a sign to every edge.

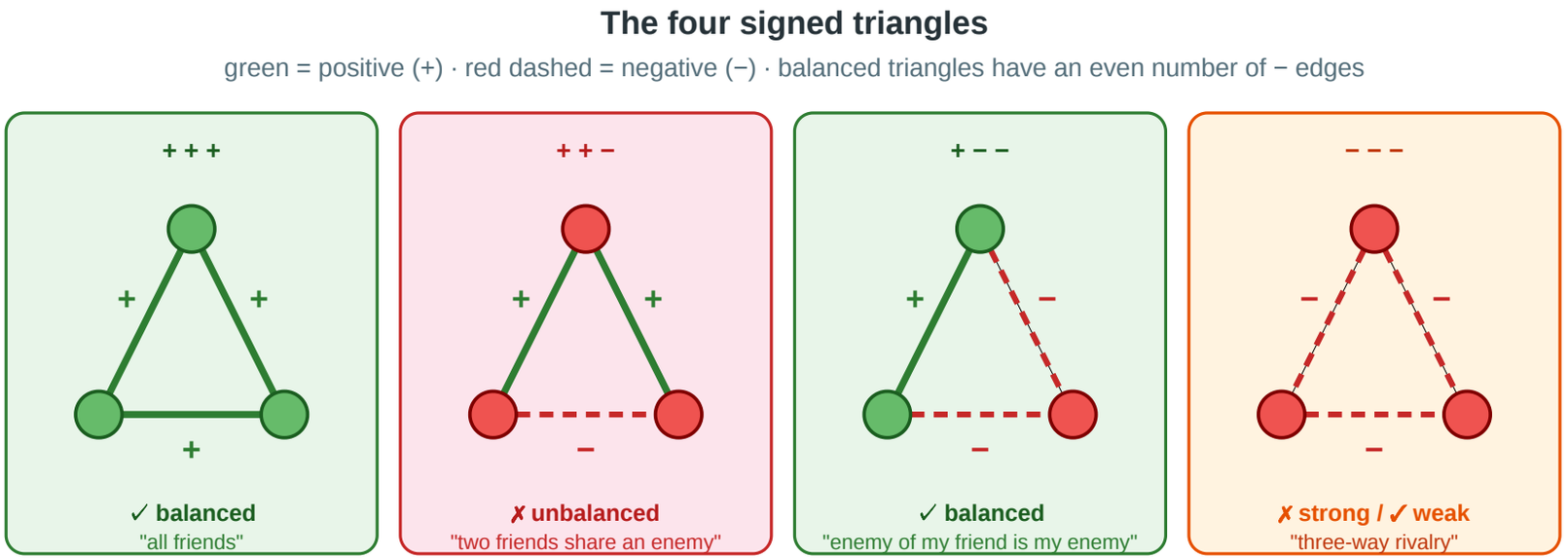
- $\sigma(e) = +$: alliance, friendship, trust
- $\sigma(e) = -$: rivalry, hostility, distrust

A **complete signed graph** has an edge (with some sign) between every pair of nodes.

Speaker notes

The completeness assumption matters: the Balance Theorem requires every pair to have a signed relationship. In the Cold War scenario, the five nations form a complete signed graph — every pair has a clear alliance or rivalry. In real social networks, most pairs have no observed edge, which is why Module 4 introduces approximate balance.

The Four Signed Triangles



A triangle is **balanced** if and only if it has an **even** number of negative edges (0 or 2). The $(+\ +\ -)$ triangle is unstable: two friends share an enemy, creating pressure for one friendship to break. The $(-\ -\ -)$ triangle is the ambiguous case — stable under weak balance but not strong.

Speaker notes

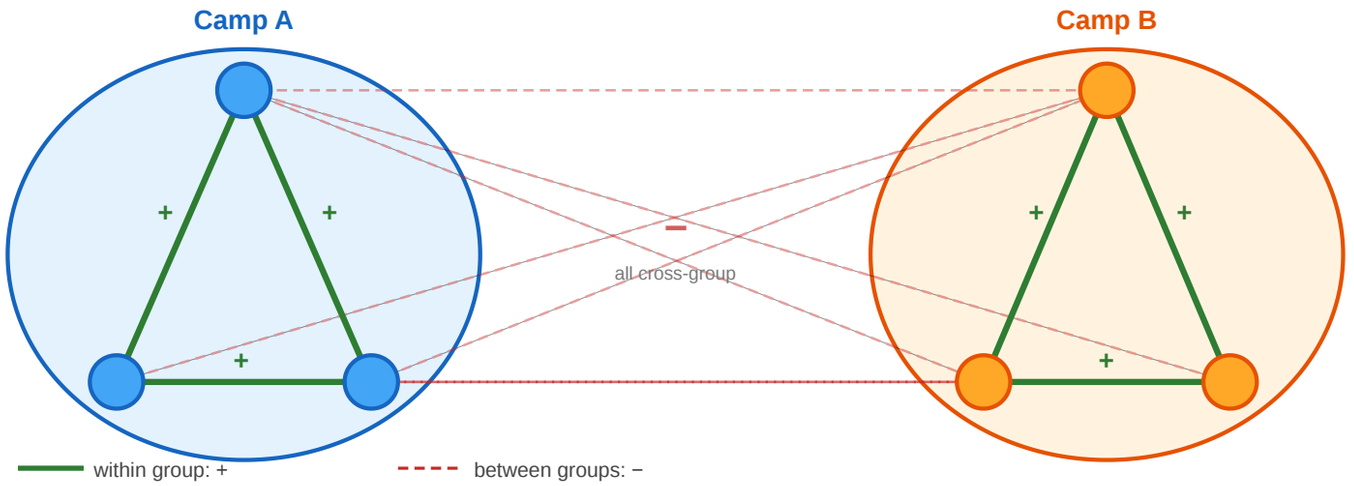
The even-number rule is the fastest way to classify triangles. The psychological interpretations are: $(+\ +\ +)$ "all friends" — no tension; $(+\ +\ -)$ "two friends disagree about a third" — tension, someone must change; $(+\ -\ -)$ "the enemy of my friend is my enemy" — stable; $(-\ -\ -)$ "three-way rivalry" — everyone would benefit from allying with *someone*, but who? The last case is what distinguishes strong from weak balance.

The Structural Balance Theorem

Structural Balance Theorem (Cartwright & Harary, 1956). A complete signed graph (G, σ) is balanced — i.e., every triangle has an even number of negative edges — if and only if the nodes can be partitioned into **at most two** groups such that:

- every edge *within* a group is positive
- every edge *between* groups is negative

In other words: **balance implies two camps**.

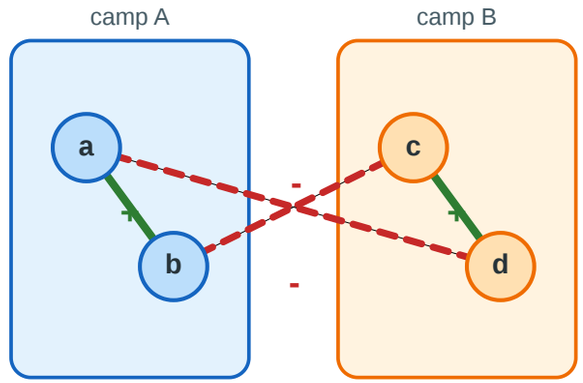


Speaker notes

This is one of the cleanest results in network science. A local triangle constraint produces a rigid global structure with only two possible shapes: one all-positive group (if everyone is friends) or two internally friendly groups with mutual hostility. There is no third option. The Cold War pre-1960 panel is the canonical illustration: {USA, UK, France} vs. {USSR, China}.

Proof Sketch

1. Camps -> balanced triangles



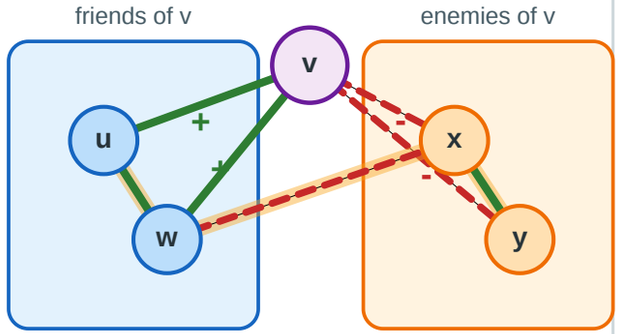
inside one camp

0 negative edges

split 2 + 1

2 negative edges

2. Balanced triangles -> camps



Triangles with v force all remaining signs:

within each side: +

across the two sides: -

Camps → balance: within-camp edges are positive and cross-camp edges are negative, so every triangle has either 0 or 2 negative edges.

Balance → camps: pick one node v. Its friends must form one positive camp, its enemies must form another positive camp, and all edges across the two camps are forced to be negative.

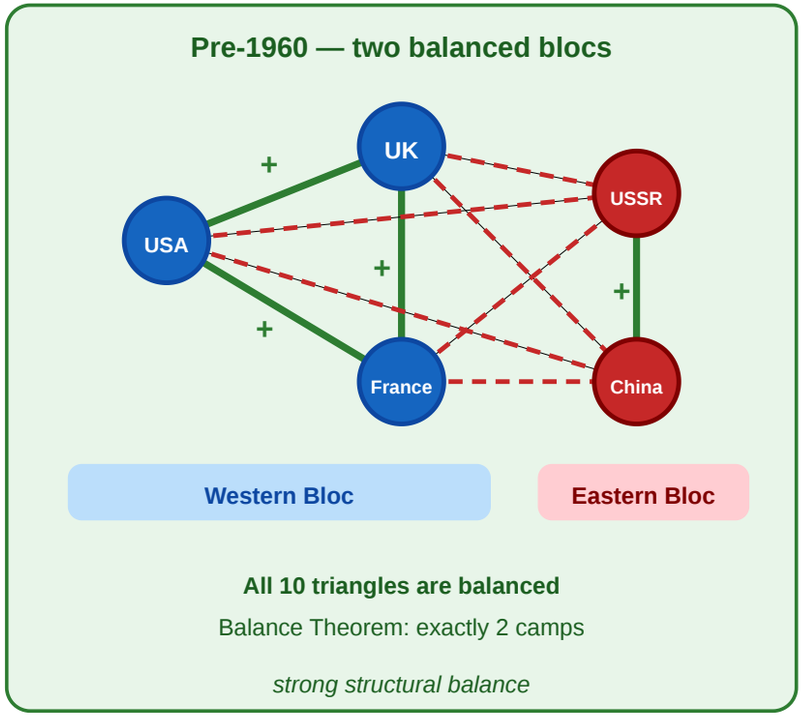
Speaker notes

The proof of direction 2 is worth walking through explicitly because it shows *why* the local rule cascades globally. Once you fix one node's partition, every other node's assignment is forced. The proof also reveals the key assumption: *completeness*. If some edges are missing, the cascade does not work — which is exactly the structural gap we flagged in the introduction.

Back to the Cold War

In the pre-1960 panel: **all 10 triangles** in K_5 are balanced.

Triangle	Signs	Count of −	Balanced?
USA–UK–France	$+\ +\ +$	0	✓
USA–UK–USSR	$+\ -\ -$	2	✓
USA–UK–China	$+\ -\ -$	2	✓
USA–France–USSR	$+\ -\ -$	2	✓
USA–France–China	$+\ -\ -$	2	✓
USA–USSR–China	$-\ -\ +$	2	✓
UK–France–USSR	$+\ -\ -$	2	✓
UK–France–China	$+\ -\ -$	2	✓



Triangle	Signs	Count of −	Balanced?
UK–USSR–China	$-\ -\ +$	2	✓
France–USSR–China	$-\ -\ +$	2	✓

The theorem correctly predicts two camps: {USA, UK, France} and {USSR, China}.

Speaker notes

Walking through every triangle on a small example makes the theorem concrete. Students can verify each row, and the pattern becomes visible: the only triangles with 0 negative edges are all-Western; all others have exactly 2 negative edges (one cross-bloc edge per Western pair). The count confirms the prediction mechanically.

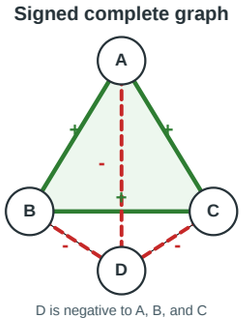
Takeaway

The Structural Balance Theorem is a rare example of a local constraint (triangle psychology) producing a clean global prediction (two camps). A single triangle rule, applied consistently across a complete signed graph, locks the entire network into a rigid two-bloc structure.

Quick Check

Consider four nations with the signed graph shown on the right.

1. List all four triangles and classify each as balanced or unbalanced.
2. Is the graph balanced? If so, what are the two camps?
3. If we flip the C–D edge from − to +, which triangle(s) become unbalanced?



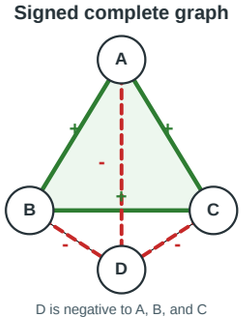
Speaker notes

Give them 5 minutes. (1) ABC: +++, balanced. ABD: +−−, balanced. ACD: +−−, balanced. BCD: +−−, balanced. (2) All balanced → yes. Camps: {A,B,C} vs {D}. (3) Flipping C–D to + makes triangle BCD become ++− (unbalanced) and ACD become ++− (unbalanced), while ABC stays +++ and ABD stays +−−.

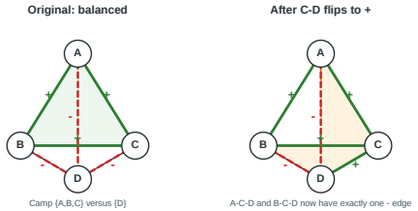
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Quick Check — Solution



1. Before the flip: all four triangles are balanced.

- A-B-C has 0 negative edges.
- A-B-D, A-C-D, and B-C-D each have 2 negative edges.

2. All balanced → **yes**. Camps: $\{A, B, C\}$ vs $\{D\}$.

3. Flipping C–D to + makes:

- A-C-D → one negative edge ✗
- B-C-D → one negative edge ✗

Weak Structural Balance

Guiding Question: What happens to the two-camps theorem when three-way rivalries are allowed?

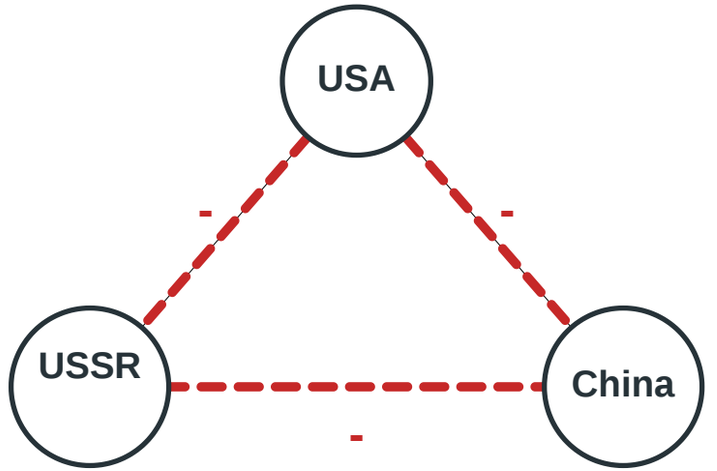
Speaker notes

The Sino-Soviet split in the Cold War scenario motivates this module directly. After 1969, the triangle USA–USSR–China is all-negative (−−−). Under strong balance this is forbidden; under weak balance it is allowed. The global consequence: instead of exactly two camps, we get *multiple* camps.

The Problem with All-Negative Triangles

In the post-1969 Cold War panel, the triangle USA–USSR–China has three negative edges. Under strong balance, this is **unbalanced** — the theorem says the graph cannot split into two camps. But historically, the world *did* function with three separate power blocs: {USA, UK, France}, {USSR}, {China}.

Post-1969: all-negative triangle



Strong balance: forbidden
Weak balance: allowed as three separate camps

Speaker notes

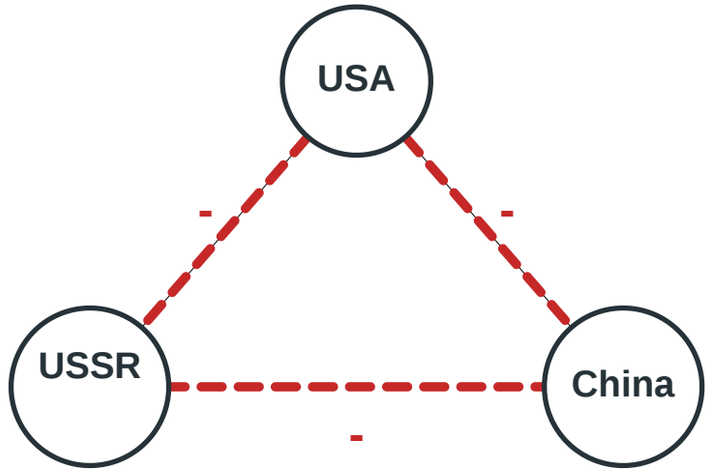
The psychological distinction is important: $(+\ +\ -)$ creates active dissonance within the triangle (your friend's enemy is your friend), while $(-\ -\ -)$ creates no such contradiction — nobody expects their enemy's enemy to be their friend. Davis (1967) formalized this distinction into weak structural balance.

The Problem with All-Negative Triangles

In the post-1969 Cold War panel, the triangle USA–USSR–China has three negative edges. Under strong balance, this is **unbalanced** — the theorem says the graph cannot split into two camps. But historically, the world *did* function with three separate power blocs: {USA, UK, France}, {USSR}, {China}.

The issue is psychological: a three-way rivalry may be *unstable* (each party would benefit from allying with someone), but it is not *contradictory* in the way that $(+ + -)$ is. Two friends disagreeing about a third face direct tension; three mutual enemies merely face *opportunity*.

Post-1969: all-negative triangle



Strong balance: forbidden

Weak balance: allowed as three separate camps

Weak Balance — Definition

> **Weak Structural Balance (Davis, 1967).** A complete signed graph is **weakly balanced** if it contains no triangle with exactly **one** negative edge $(+\ +\ -)$. All-negative triangles $(-\ -\ -)$ are permitted.

A **complete signed graph** assigns a + or − label to every edge between every pair of nodes — every pair is connected by exactly one signed edge.

Speaker notes

The relaxation is small — permit one additional triangle type — but the global consequence is large: instead of ≤2 camps, we can have *any number* of internally friendly, mutually hostile coalitions. The Cold War after 1969 illustrates $k = 3$: Western Bloc, Soviet Bloc, China. This is a powerful example of how changing one local rule changes the entire global picture.

Weak Balance — Definition

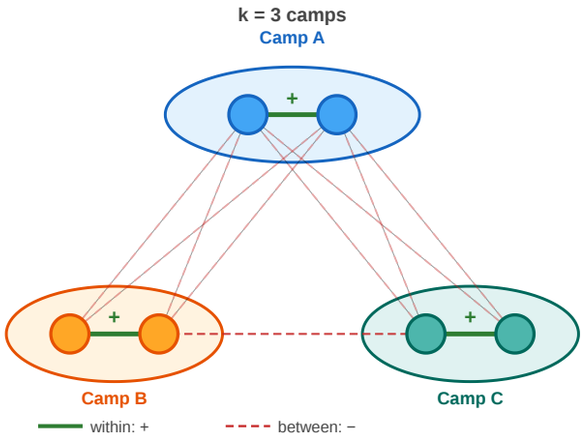
> **Weak Structural Balance (Davis, 1967).** A complete signed graph is **weakly balanced** if it contains no triangle with exactly **one** negative edge $(+ + -)$. All-negative triangles $(- - -)$ are permitted.

A **complete signed graph** assigns a + or − label to every edge between every pair of nodes — every pair is connected by exactly one signed edge.

Weak Balance Theorem. A complete signed graph is weakly balanced if and only if the nodes can be partitioned into $k \geq 1$ groups such that:

- every edge *within* a group is positive
- every edge *between* groups is negative

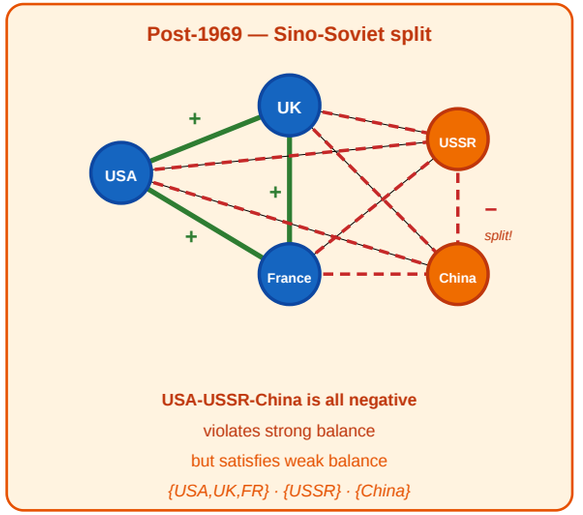
(Strong balance is the special case $k \leq 2$.)



Back to the Cold War — Post-1969

After the Sino-Soviet split, the five-nation graph has:

- $(+\ +\ +)$ triangles: USA–UK–France ✓
- $(+\ -\ -)$ triangles: all Western-pair + one Eastern node (6 triangles) ✓
- $(-\ -\ -)$ triangles: USA–USSR–China, UK–USSR–China, France–USSR–China ✓ under weak balance



Speaker notes

The check is now easy: students just need to confirm no $(+\ +\ -)$ triangle exists. The three all-negative triangles are exactly the ones that strong balance forbids but weak balance permits. The result: three camps instead of two, matching the historical reality of a multipolar Cold War after 1969.

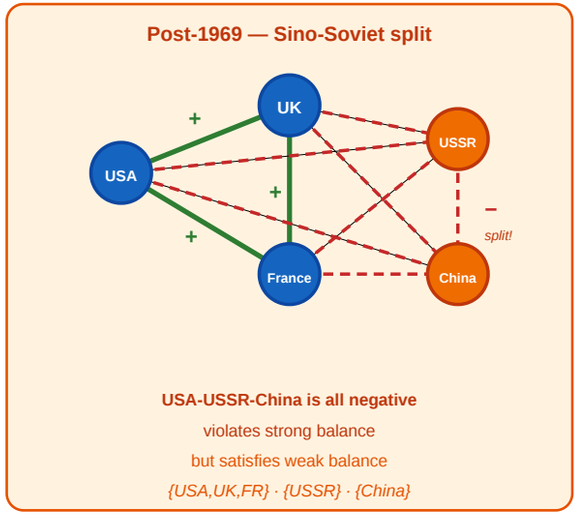
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- $(+\ -\ -)$ triangles: all Western-pair + one Eastern node (6 triangles) ✓
- $(-\ -\ -)$ triangles: USA–USSR–China, UK–USSR–China, France–USSR–China ✓ under weak balance

No $(+\ +\ -)$ triangles exist → **weakly balanced**.

Partition into 3 camps: $\{\mathrm{USA}, \mathrm{UK}, \mathrm{France}\}$, $\{\mathrm{USSR}\}$, $\{\mathrm{China}\}$.



Takeaway

Weak balance replaces the two-bloc theorem with a k-coalition theorem. Permitting all-negative triangles allows multiple hostile camps — a better model for real-world multipolar structures like the post-1969 Cold War.

Empirical Test: Signed Networks in Social Media

Guiding Question: In real signed networks with millions of edges, does balance actually hold?

Speaker notes

This module provides the empirical anchor for L06, following the same pattern as Kossinets-Watts (L03), Zachary (L04), and Christakis-Fowler (L05). The study is Leskovec, Huttenlocher & Kleinberg (2010), which analyzed three large online signed networks and found that triangle-level balance holds remarkably well — but with important caveats.

The Study

Leskovec, Huttenlocher & Kleinberg (2010), *Predicting Positive and Negative Links in Online Social Networks*, **Proc. 19th International Conference on World Wide Web (WWW), 641–650**.

They analyzed three large signed networks:

Dataset	What is a node?	What is a directed signed edge?	Size
Epinions	users of a product-review site	user u marks user v as **trusted** (+) or **distrusted** (−)	119 217 users; 841 200 edges; 85.0% +
Slashdot	users of the technology-news site	user u tags user v as **friend** (+) or **foe** (−) in Slashdot Zoo	82 144 users; 549 202 edges; 77.4% +
Wikipedia	users involved in admin elections	voter u casts **support** (+) or **oppose** (−) vote on candidate v for adminship	7 118 users; 103 747 votes; 78.7% +

Wikipedia here is an **election/voting network**, not an article-editing or friendship network: edges run from voters to admin candidates across 2 794 administrator elections.

Speaker notes

These are among the largest signed networks studied in the paper. The important detail is that all signs are explicit user actions, not inferred sentiment. Epinions covers trust and distrust links on a product-review site from the site's beginning in 1999 until August 12, 2003. Slashdot uses the Slashdot Zoo feature, crawled in February 2009, where users could label others as friends or foes. Wikipedia is different: it is not a general social graph or an article co-editing graph. It is a directed voting graph extracted from Wikipedia administrator election histories in a January 2008 dump: voters cast support or oppose votes on candidates. The paper reports 2,794 elections, 1,235 successful promotions, and 1,559 unsuccessful ones. All three datasets are directed signed graphs, although later balance analyses often simplify direction to compare with undirected balance theory.

What They Found

Triangle-level balance is remarkably strong:

Triangle type	Expected (random signs)	Observed (Epinions)
+ + +	~12.5%	~47%
+ + −	~37.5%	**~8%**
+ − −	~37.5%	~37%
− − −	~12.5%	~8%

Speaker notes

The random baseline assumes each edge is independently + or − with the observed overall positive rate. The deviation from this baseline is the evidence for balance. The key number: $(+\ +\ -)$ is suppressed by a factor of ~5×. This is the strongest possible evidence that local balance constraints shape global structure in real signed networks. The $(-\ -\ -)$ underrepresentation is also noteworthy — it falls between the strong and weak balance predictions.

What They Found

Triangle-level balance is remarkably strong:

Triangle type	Expected (random signs)	Observed (Epinions)
$+ + +$	~12.5%	~47%
$+ + -$	~37.5%	**~8%**
$+ - -$	~37.5%	~37%
$- - -$	~12.5%	~8%

The $(+ + -)$ type — the one both strong and weak balance forbid — is **massively underrepresented** (~8% vs. 37.5% expected). The balanced types $(+ + +)$ and $(+ - -)$ are overrepresented. Balance is not perfect, but the skew is dramatic.

What the Study Could Not See

1. Sparse, not complete. The Balance Theorem assumes a complete signed graph. These networks are extremely sparse — most pairs have no edge. The theorem's prediction (two or k camps) does not directly apply; the study tests triangle statistics, not the global partition.

2. Directed signs. On Epinions, trust and distrust are *directed* — A can trust B without B trusting A. The balance framework assumes undirected signs. Directed balance theory exists (Leskovec et al. explore it) but is less clean.

3. Missing negative edges. People are less likely to *create* a negative edge than to simply *not interact*. The observed negative edges are a biased sample: they represent actively expressed hostility, not indifference. The "true" signed graph — if every pair had a relationship — might look different.

Speaker notes

This follows the L03–L05 pattern: celebrate the result, then identify blind spots. The sparsity issue is the most fundamental: the Balance Theorem is a statement about complete graphs, and no real network is complete. The study's response — test triangle statistics rather than global partition — is a pragmatic substitute, but it leaves open whether the *global* structure is consistent with balance. Module 4 introduces tools for addressing this gap.

Takeaway

Leskovec et al. (2010) showed that balance holds at the triangle level in large online signed networks: the forbidden $(+ + -)$ type is massively underrepresented. But real networks are sparse and directed, so the clean two-camp prediction of the Balance Theorem must be tested through approximate measures, not exact partition.

Incomplete and Approximate Balance

Guiding Question: How can we assess balance when the graph is incomplete or only approximately balanced?

Speaker notes

This module addresses the structural gap identified in the introduction: the Balance Theorem requires completeness, but real networks are sparse. We need tools that work on partial data. Two approaches: the cycle-based criterion (extend balance from triangles to all cycles) and the frustration index (measure distance from the nearest balanced graph).

From Triangles to Cycles

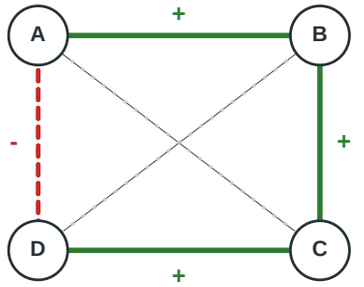
Balance was defined on triangles — but sparse graphs may have **no triangles at all**, making that definition vacuous.

Complete graph: Every pair of nodes is connected, so every path passes through a triangle. Triangles cover the entire structure — no further check is needed.

Sparse graph: Many pairs have no edge. There may be *no triangles at all*, so the triangle criterion is trivially satisfied — nothing to check. Yet longer cycles can still be unbalanced.

Cycle criterion (Harary 1953). A signed graph (G, σ) is balanced if and only if every cycle in G has an **even** number of negative edges.
On a complete graph this reduces to checking triangles. On a sparse graph it is strictly stronger.

Sparse graph: no triangles, but still unbalanced



Triangle test:
no triangles exist

Cycle test:
one negative edge
odd -> unbalanced

Counterexample. 4-cycle A–B(+)–C(+)–D(+)–A(−): no triangles → triangle criterion passes trivially. One negative edge → cycle is **unbalanced**.

Speaker notes

The key pedagogical move is the vacuousness argument: balance was defined as "every triangle has an even number of negative edges." On a sparse graph that has no triangles, this condition is trivially true — there is nothing to falsify it. But the graph can still be structurally unbalanced in the sense that a two-camp partition does not exist. The 4-cycle example is the clearest illustration: 4 nodes in a ring, no diagonals, one negative edge. Zero triangles → triangle check says balanced. But the product of signs around the only cycle is − → the graph is not balanced under the cycle criterion. Walk students through: +·+·+·(−) = −. Harary's 1953 generalization replaces the triangle criterion with the cycle criterion, which is vacuously equivalent on complete graphs (every cycle decomposes into triangles there) but strictly stronger on sparse ones.

The Frustration Index

In real data, perfect balance is rare. The natural question becomes: **how far** is the graph from balanced?

Frustration index $F(G, \sigma)$: the minimum number of edges whose sign must be flipped to make the graph balanced.

$$F(G, \sigma) = \min_{\sigma'} |\{e \in E : \sigma(e) \neq \sigma'(e)\}|$$

subject to (G, σ') being balanced.

Speaker notes

The frustration index is the signed-graph analogue of the edit distance to the nearest balanced partition. Computing it exactly is NP-hard in general (it reduces to a graph-cut problem), but good approximations exist — including spectral methods that use the signed Laplacian. The normalized version $\int$ provides a scale-independent measure that can be compared across networks.

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Normalized frustration: $f = F/|E|$. If $f \approx 0$, the graph is near-balanced; if $f \approx 0.5$, it is no more balanced than a randomly signed graph.

Computing Approximate Balance

Exact computation. Finding F exactly is NP-hard (equivalent to a minimum-weight graph cut on a related unsigned graph). On small graphs, integer programming or brute force works; on large graphs, we need heuristics.

Spectral approach. The **signed Laplacian** $L_\sigma = D - A_\sigma$ (where A_σ has entries $+1$ or -1) has the property that (G, σ) is balanced iff the smallest eigenvalue $\lambda_1(L_\sigma) = 0$. The magnitude of λ_1 quantifies how far the graph is from balanced.

Practical rule of thumb. For large networks, count the fraction of balanced triangles ($T_{\text{bal}}/T_{\text{total}}$) as a fast proxy. Leskovec et al.'s Epinions data had $T_{\text{bal}}/T_{\text{total}} \approx 0.92$.

What is the Laplacian? For an unsigned graph, the Laplacian $L = D - A$ encodes structure: D is the diagonal degree matrix and A is the adjacency matrix. Its eigenvalues measure connectivity — $\lambda_1 = 0$ iff the graph is connected. The **signed Laplacian** replaces A with A_σ (entries $+1$ for positive edges, -1 for negative edges). Now $\lambda_1 = 0$ iff the graph is balanced — balance plays the same role that connectivity plays in the unsigned case.

Speaker notes

The spectral approach connects to L04's eigenvector centrality: both use properties of the graph's matrix representation. The signed Laplacian is the natural extension of the unsigned Laplacian to signed graphs, and its spectrum encodes balance information just as the unsigned Laplacian's spectrum encodes connectivity information. For large networks, the balanced-triangle fraction is the cheapest informative statistic — it is what Leskovec et al. actually computed.

Quick Check

A signed graph on 5 nodes has 7 edges. You compute that 8 out of 10 triangles are balanced and 2 are unbalanced (both of type $+ + -$).

1. Is the graph balanced? Why or why not?
2. Can you conclude that the frustration index is exactly 2?
3. What additional information would you need to determine the exact frustration index?

Speaker notes

Give them 4 minutes. (1) No — balanced requires *all* triangles (and all cycles) to be balanced. Two unbalanced triangles means the graph is not balanced. (2) No — the frustration index counts *edge flips*, not unbalanced triangles. Two unbalanced triangles might share an edge; flipping that one edge could fix both. So $F \leq 2$ but could be 1. (3) You would need the full edge list and sign assignment to determine which specific edges, when flipped, resolve all imbalances with the fewest changes.

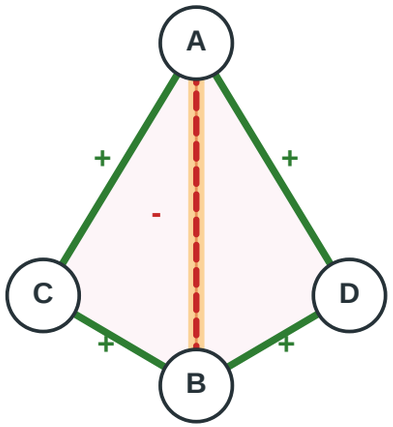
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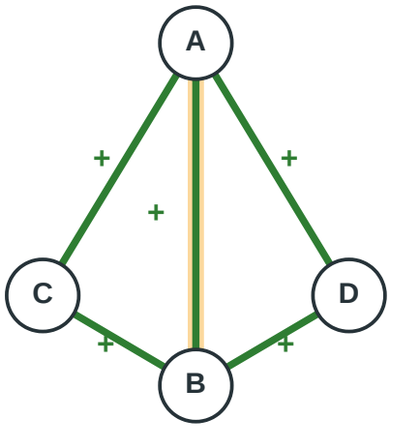
Quick Check — Solution

Before: two bad triangles



A-C-B and A-D-B are both ++-

Flip one shared edge



One flip fixes both triangles: F can be 1

1. Not balanced. Two unbalanced triangles violate the "every triangle must be balanced" condition.

2. No. Frustration index counts *edge flips*, not unbalanced triangles. The two bad triangles might share an edge — flipping that one shared edge could fix

both. So $F \leq 2$, but F could be as low as 1.

3. You need the full graph — which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

Takeaway

> Real signed networks are sparse and imperfect. The frustration index and the balanced-triangle fraction are the practical tools for measuring how close a network is to the balanced ideal — replacing the all-or-nothing theorem with a quantitative diagnostic.

Wrap-Up — Lecture 06

L06 added **signs** to edges. Once ties can be positive or negative, local triangle patterns constrain the possible global structure.

Question	L06 answer
Which signed triangles are stable?	Strong balance allows $+\ +\ +$ and $+\ -\ -$; weak balance also allows $-\ -\ -$.
What does local balance imply globally?	Strong balance forces at most two camps; weak balance allows k hostile camps.
Do real online networks show balance?	Signed social-media data strongly suppresses the forbidden $+\ +\ -$ triangle.
Why is the theorem not enough?	Real signed networks are sparse, directed, and noisy; exact complete-graph balance rarely applies.
How do we measure approximate balance?	Use cycle parity, frustration index, signed Laplacian ideas, and balanced-triangle fractions.

The central lesson: signed networks turn "who is connected?" into "what kind of relation is this?" The sign pattern itself becomes structure.

Reading

Chapter 5 of Easley & Kleinberg (2010) covers structural balance theory, weak balance, and applications. Heider (1946) and Cartwright & Harary (1956) are the original sources. Leskovec, Huttenlocher & Kleinberg (2010) is the empirical study on signed social networks.

Speaker notes

Close by returning to the L06 arc only. The lecture started with signed edges, classified local triangles, proved that complete balanced graphs collapse into camps, relaxed the rule to weak balance, then asked what remains in sparse empirical data. The key conceptual shift is that signs are not decorations on edges: they impose constraints on the whole graph. The empirical and approximate-balance sections are there because real signed networks violate the assumptions of the clean theorem.